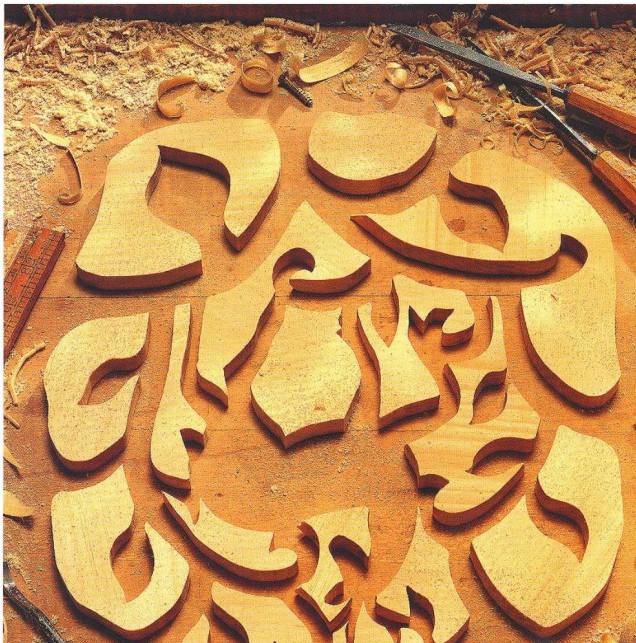


Lambertian model of reflectance I: shape from shading and photometric stereo

Bump or dip?



What sticks out?

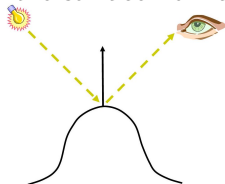


Light field

- ▶ Rays travel from sources to objects
- ▶ There they are either absorbed or reflected
- ▶ Energy of light decreases with distance and number of bounces
- ▶ Camera is designed to capture the set of rays that travel through the focal center
- ▶ We will assume the camera response is linear

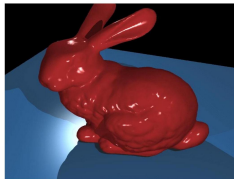
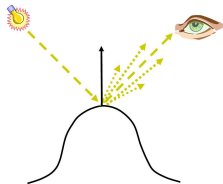
Specular reflectance (mirror)

- ▶ When a surface is smooth light reflects in the opposite direction of the surface normal



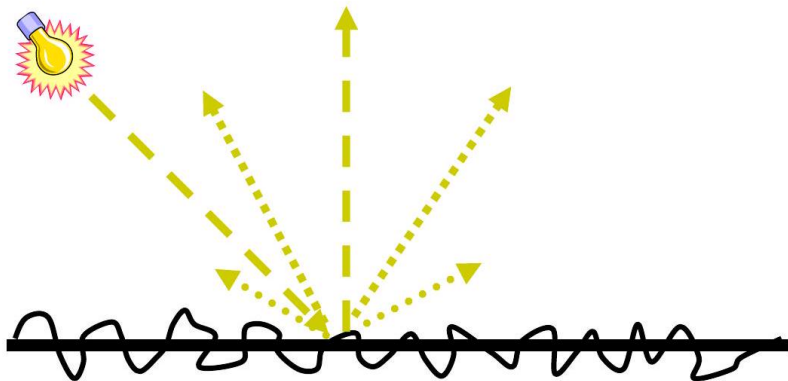
Specular reflectance

- ▶ When a surface is slightly rough the reflected light will fall off around the specular direction



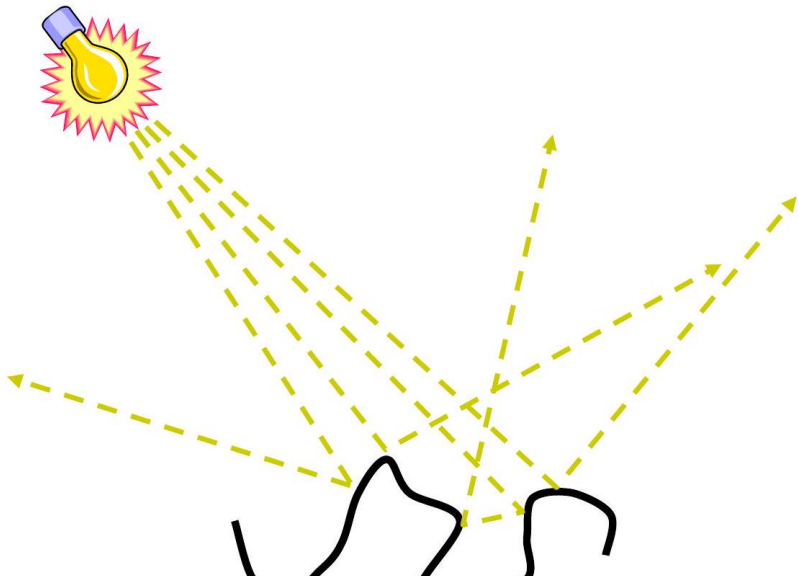
Diffuse reflectance

- ▶ When the surface is very rough light may be reflected equally in all directions



Diffuse reflectance

- ▶ When the surface is very rough light may be reflected equally in all directions

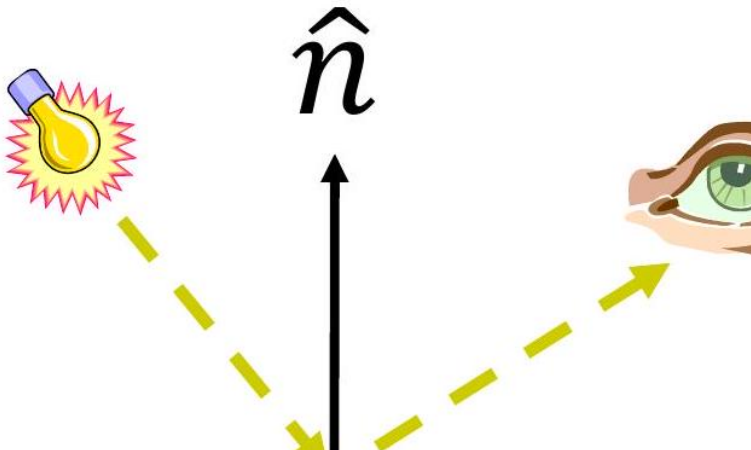


BRDF

- Bidirectional Reflectance Distribution Function

$$f(\theta_{\text{in}}, \phi_{\text{in}}; \theta_{\text{out}}, \phi_{\text{out}})$$

- Specifies for a unit of incoming light in a direction $(\theta_{\text{in}}, \phi_{\text{in}})$ how much light will be reflected in a direction $(\theta_{\text{out}}, \phi_{\text{out}})$

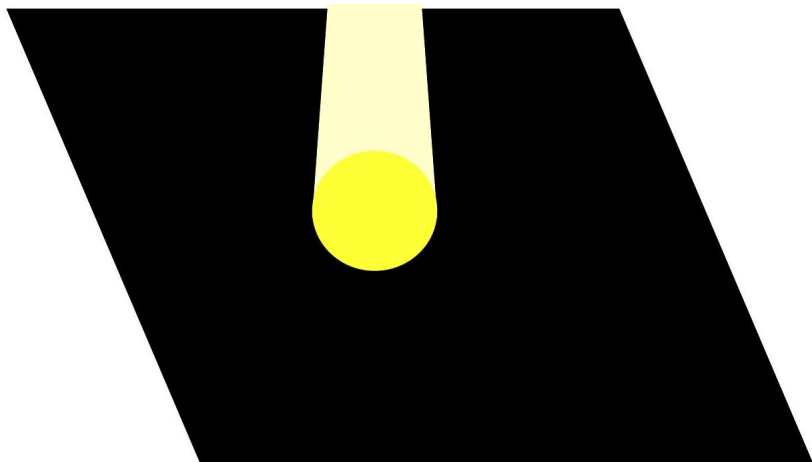


BRDF

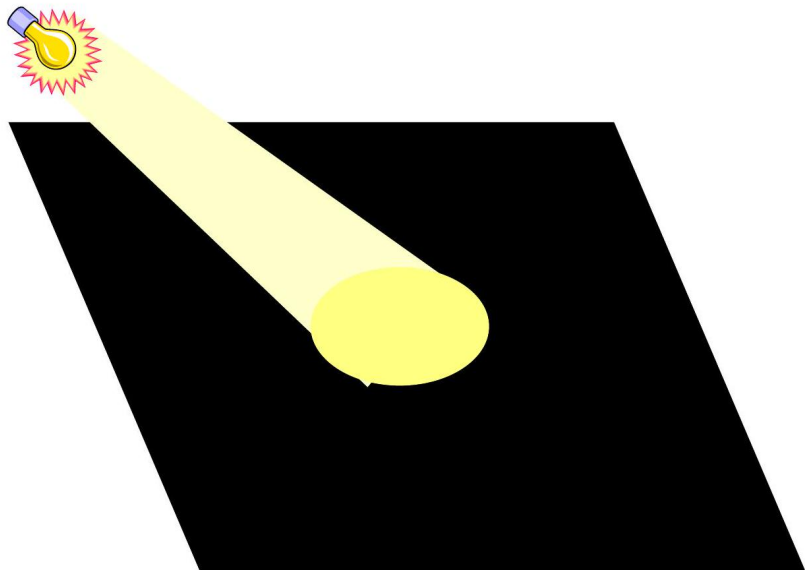


Light from front Light from back

Lambertian reflectance



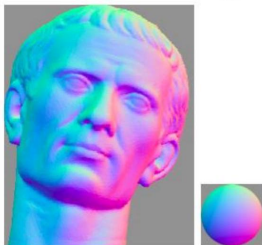
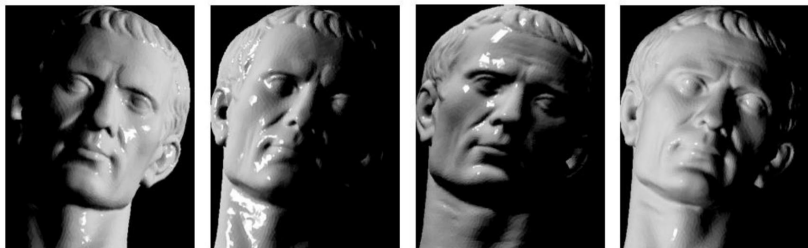
Lambertian reflectance



Bas-relief



Photometric stereo with matrix completion (Wu et al.)



Example

